

**DRAFT**

**DEGREE, DEFECT AND SEPARATORS IN  
 $K_7^-$ -MINOR-FREE CRITICAL GRAPHS**

CAVIT ERGINSOY

**ABSTRACT.** Let  $G$  be a minor-minimal non-six-colourable graph with no  $K_7^-$  minor. For each  $i$ , let  $n_i$  count its degree- $i$  vertices, and put  $\tau = \sum_{i \geq 10} (i - 9)n_i$ . We prove that  $G$  is seven-connected, has minimum degree eight, contains no  $K_5$  subgraph, and satisfies  $n_8 \geq 27 + \tau$ . Every degree-eight vertex is incident with an edge lying in at most three triangles, and deleting any seven-vertex cut leaves exactly two components. The bound on  $n_8$  follows from the broader statement that every  $r$ -connected  $K_7^-$ -minor-free graph with  $r \geq 6$  and at least four times as many edges as vertices has defect  $9|V| - 2|E| \geq 20 + r$ . This statement uses a computer-assisted finite lemma on nine- and ten-vertex quotients. We also give computation-free proofs of  $n_8 \geq 26 + \tau$ , the incident-edge conclusion and the separator conclusion.

## 1. INTRODUCTION

Hadwiger's conjecture states that every graph of chromatic number  $t$  contains a  $K_t$  minor [4]. It is known for  $t \leq 6$ , with the last case due to Robertson, Seymour and Thomas [11], and remains open for  $t \geq 7$ . Write  $K_t^-$  for the graph obtained from  $K_t$  by deleting one edge, and  $K_7^\vee$  for the graph obtained from  $K_7$  by deleting two incident edges. Albar proved that every  $K_7^-$ -minor-free graph is seven-colourable [1, Theorem 1]. Norin and Totschnig proved that every  $K_7^\vee$ -minor-free graph is six-colourable [10, Theorem 4]. They asked whether  $K_7^-$  may replace  $K_7^\vee$  in that statement [10, Conjecture 21].

A graph is *minor-minimal non-six-colourable* if it is not six-colourable and every proper minor is six-colourable. The following theorem describes such a graph under the exclusion in Conjecture 21.

**Theorem 1.1.** *Let  $G$  be a minor-minimal non-six-colourable  $K_7^-$ -minor-free graph. For each  $i$ , let  $n_i$  denote the number of vertices of degree  $i$ , and put  $\tau = \sum_{i \geq 10} (i - 9)n_i$ . Then  $G$  is seven-connected and*

$$\delta(G) \geq 8, \quad |E(G)| \geq 4|V(G)|.$$

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Moreover,  $G$  contains no  $K_5$  subgraph,

$$n_8 \geq 27 + \tau,$$

and the neighbourhood of every degree-eight vertex is  $K_4$ -free and has independence number three. Every such vertex is incident with an edge lying in at most three triangles. Every seven-vertex cut of  $G$  leaves exactly two components.

The bound  $n_8 \geq 27 + \tau$  uses the finite lemma 7.2. All the other conclusions, together with the weaker bound  $n_8 \geq 26 + \tau$ , have computation-free proofs.

These are necessary conditions for a counterexample to Conjecture 21. A sufficient open assertion is that every seven-connected graph  $H$  satisfying  $|E(H)| \geq 4|V(H)|$  contains a  $K_7^-$  minor. The bound on  $n_8$  follows from the next theorem.

**Theorem 1.2.** *Let  $r \geq 6$ , and let  $H$  be an  $r$ -connected  $K_7^-$ -minor-free graph with  $|E(H)| \geq 4|V(H)|$ . Then*

$$9|V(H)| - 2|E(H)| \geq 20 + r.$$

*The proof uses the finite lemma 7.2.*

The proofs use linked cliques and rooted models to exclude vertices of degree seven and  $K_5$  subgraphs. Section 6 then gives the computation-free bound  $n_8 \geq 26 + \tau$ . Theorem 1.2 uses a density-preserving contraction and induction on connectivity (Section 7). The separator proof aligns boundary colourings obtained from proper minors and extends precolourings over planar graphs (Section 8).

Mayer obtained a finite catalogue of candidate neighbourhood graphs for degree-seven vertices in six-contraction-critical graphs [9]. In the minor-minimal  $K_7^\vee$ -minor-free setting, Norin and Totschnig proved that every degree-seven vertex belongs to a  $K_5$  contained in its closed neighbourhood [10, Claim 4.4]. We first determine the neighbourhood under  $K_7^-$  exclusion, then rule out degree seven.

Rolek and Song proved the next result when  $r = 6$  [12, Lemma 1.9]. We shall use the uniform form with  $r = 5$ .

**Theorem 1.3.** *For every  $r \geq 2$ , if  $G$  is an  $(r+1)$ -connected graph containing two distinct  $K_r$  subgraphs, then  $K_{r+2}^- \preccurlyeq G$ .*

The connectivity hypothesis is sharp. The graph  $K_{r+1}$  is  $r$ -connected and contains two distinct  $K_r$  subgraphs, but has no  $K_{r+2}^-$  minor.

The following rooted closure will exclude the remaining degree-seven neighbourhood.

**Theorem 1.4.** *Let  $G$  be a six-connected graph of order  $n$ , let  $v \in V(G)$  have degree  $d \geq 5$ , and suppose that  $G[N(v)]$  contains a  $K_4$  subgraph. If*

$$|E(G)| \geq 4n + d - 13,$$

*then  $K_7^- \preccurlyeq G$ .*

A *minor model* of a graph  $H$  in  $G$  is a family  $(B_x : x \in V(H))$  of pairwise disjoint connected vertex sets such that  $B_x$  and  $B_y$  are adjacent whenever  $xy \in E(H)$ . A model is *rooted* at prescribed vertices when its branch sets contain them one-to-one. We write  $H \preceq G$  when  $H$  is a minor of  $G$ . All graphs are finite and simple.

## 2. TWO LINKED CLIQUES

**Lemma 2.1.** *Let  $r \geq 2$ . If an  $(r+1)$ -connected graph  $H$  contains two vertex-disjoint  $K_r$  subgraphs, then  $K_{r+2}^- \preceq H$ . The branch sets can be chosen to meet the union of the two cliques.*

*Proof.* Write the two cliques as

$$A = \{a_1, \dots, a_r\}, \quad B = \{b_1, \dots, b_r\}.$$

By Menger's theorem there are  $r$  pairwise vertex-disjoint  $A$  to  $B$  paths whose open interiors avoid  $A \cup B$ . Relabel them as paths  $P_i$  from  $a_i$  to  $b_i$ , and put  $U = \bigcup_i V(P_i)$ .

Call a path  $Q$  a *usable cross-path* if its ends lie on distinct paths  $P_i, P_j$ , its open interior avoids  $U$ , and its ends are not both  $A$ -ends or both  $B$ -ends. Suppose first that  $Q$  exists. Split each of  $P_i$  and  $P_j$  across an edge into a nonempty  $A$ -prefix and a nonempty  $B$ -suffix, choosing the split so that the ends of  $Q$  belong to pieces of opposite kinds. Absorb the open interior of  $Q$  into the piece containing one end. The four resulting pieces are connected and disjoint. The two  $A$ -pieces are adjacent through  $A$ , the two  $B$ -pieces are adjacent through  $B$ , the two pieces from each original path are adjacent across its splitting edge, and  $Q$  supplies one of the two remaining cross-adjacencies. Thus at most one adjacency among the four pieces is missing. Each unsplit path is adjacent to every split piece through its two clique ends, and the unsplit paths are pairwise adjacent. These  $r+2$  sets form a  $K_{r+2}^-$  model.

Suppose now that no usable cross-path exists. Let  $C$  be a component of  $H - U$ . If  $C$  has neighbours on two distinct paths, a path through  $C$  between two such attachments is unusable only when both attachments are  $A$ -ends or both are  $B$ -ends. Comparing all pairs of attachments shows that all neighbours of  $C$  are  $A$ -ends or all are  $B$ -ends. Hence  $|N_H(C)| \leq r$ , contrary to  $(r+1)$ -connectivity. Every component of  $H - U$  therefore attaches to only one  $P_i$ .

For each  $i$ , let  $W_i$  consist of the open interior of  $P_i$  and the components of  $H - U$  attached to  $P_i$ . There is no edge from  $W_i$  to  $P_j$  for  $j \neq i$ : an edge from the path interior would be a usable cross-path, and an edge from a component would contradict its unique attachment. Hence

$$N_H(W_i) \subseteq \{a_i, b_i\}.$$

Since  $r \geq 2$ , a nonempty  $W_i$  would give a vertex cut of order at most two. Thus every  $W_i$  is empty. It follows that  $H$  consists of  $A$ ,  $B$  and the matching

edges  $a_i b_i$ : any other edge between  $A$  and  $B$  would be usable. This graph has minimum degree  $r$ , again contradicting  $(r+1)$ -connectivity.  $\square$

*Proof of Theorem 1.3.* Let  $L_1, L_2$  be distinct  $K_r$  subgraphs and put  $s = |V(L_1) \cap V(L_2)|$ . If  $s \leq r-2$ , delete the common set  $Z$ . The graph  $G-Z$  is  $(r+1-s)$ -connected and contains the two disjoint  $K_{r-s}$  subgraphs  $L_1-Z$  and  $L_2-Z$ . Lemma 2.1 gives a  $K_{r-s+2}^-$  model in which every branch set contains a vertex of  $(L_1 \cup L_2) - Z$ . The vertices of  $Z$ , taken as singleton branch sets, are complete to this model and to one another. They complete a  $K_{r+2}^-$  model.

It remains that  $s = r-1$ . The union  $X = V(L_1) \cup V(L_2)$  induces a graph containing  $K_{r+1}^-$ . Since an  $(r+1)$ -connected graph has at least  $r+2$  vertices,  $G-X$  is nonempty. Let  $C$  be a component of  $G-X$ . If  $C$  missed a vertex of  $X$ , then  $N_G(C)$  would be a cut of order at most  $r$ . Thus  $C$  is adjacent to every vertex of  $X$ . The vertices of  $X$  as singleton branch sets, together with  $C$ , form a  $K_{r+2}^-$  model.  $\square$

**Corollary 2.2.** *Every six-connected  $K_7^-$ -minor-free graph contains at most one  $K_5$  subgraph.*

### 3. DEGREE-SEVEN NEIGHBOURHOODS

A graph is  $k$ -contraction-critical if it has chromatic number  $k$  and every proper minor is  $(k-1)$ -colourable. We use two standard facts. A noncomplete  $k$ -contraction-critical graph is seven-connected when  $k \geq 7$  [8, Satz 6]; see also [12, Theorem 1.8]. If  $G$  is  $k$ -contraction-critical and  $v \in V(G)$ , then

$$(3.1) \quad \alpha(G[N(v)]) \leq d(v) - k + 2.$$

This inequality goes back to Dirac [2]; we use the formulation in [12, Lemma 1.6(i)].

We also use the following specialisation of Theorem 7 and property (\*) of Kriesell and Mohr [7, Theorem 7]. Let a graph  $J$  be properly coloured with five colours, and choose one root from each colour class. Let  $D$  be a graph on the five roots with at most six edges. If the ends of every edge  $xy \in E(D)$  lie in the same bichromatic component of  $J$  on their two colours, then  $J$  contains pairwise disjoint connected sets, one containing each root, that are adjacent for every edge of  $D$ .

**Lemma 3.1.** *Let  $G$  be a seven-chromatic graph such that every proper minor is six-colourable and  $K_7^- \not\leq G$ . Let  $v$  have degree seven, put  $H = G[N(v)]$ , and let  $ab$  be a nonedge of  $H$ . If*

$$R = N(v) \setminus \{a, b\},$$

*then  $G - v - \{a, b\}$  contains a  $K_5$  minor rooted at  $R$ .*

*Proof.* Contract the two edges  $va$  and  $vb$  to one vertex. A six-colouring of the resulting proper minor induces a six-colouring  $\varphi$  of  $G-v$  in which  $a$  and  $b$  have one common colour, say 0. Every vertex of  $R$  avoids colour 0. The

five vertices of  $R$  use the other five colours distinctly, since a colour absent from  $N(v)$  could be assigned to  $v$ .

Put  $F = \overline{H}$ . Inequality (3.1) gives  $\alpha(H) \leq 2$ , so  $F$  is triangle-free. If  $xy \in E(F[R])$  and  $x, y$  have colours  $i, j$ , then  $x, y$  lie in the same component of the subgraph of  $G - v$  induced by colours  $i, j$ . Otherwise, interchanging  $i$  and  $j$  on the component containing  $x$  would remove colour  $i$  from  $N(v)$ , allowing that colour at  $v$ .

Delete the colour-0 class from  $G - v$ . The resulting graph is properly five-coloured, with  $R$  a transversal of its colour classes. By Mantel's theorem, the triangle-free graph  $F[R]$  has at most six edges. Apply the Kriesell and Mohr result with demand graph  $F[R]$ . It gives five disjoint connected sets rooted at  $R$  and adjacent for every nonedge of  $H[R]$ . Every other pair is joined by its edge in  $H[R]$ , so the five sets form the required rooted  $K_5$  model.  $\square$

**Theorem 3.2.** *Under the hypotheses of Lemma 3.1, every degree-seven vertex  $v$  has  $G[N(v)] \cong K_4 \dot{\cup} K_3$  and hence belongs to a  $K_5$  subgraph.*

*Proof.* Put  $H = G[N(v)]$  and  $F = \overline{H}$ . By (3.1),  $F$  is triangle-free. It is nonempty, since otherwise  $G[N(v)]$  would contain  $K_7^-$ .

Every nonisolated vertex of  $F$  has degree at least three. To see this, choose an edge  $ab \in E(F)$  and use the rooted model of Lemma 3.1 on  $R = N(v) \setminus \{a, b\}$ . If  $d_F(a) = 1$ , then  $a$  is adjacent in  $H$  to every root; the rooted model,  $\{a\}$  and  $\{v\}$  form a  $K_7$  model. If  $d_F(a) = 2$ , with other neighbour  $x$ , then triangle-freeness gives  $bx \in E(H)$ . Absorb  $b$  into the branch set rooted at  $x$ . Together with  $\{a\}$  and  $\{v\}$ , the resulting seven branch sets have at most one missing adjacency, again a contradiction.

Every nontrivial component of  $F$  therefore has minimum degree at least three and hence at least six vertices. Since  $|V(F)| = 7$ , there is one nontrivial component and at most one isolated vertex. If the component has six vertices, Mantel's theorem and its equality case give  $F \cong K_{3,3} \dot{\cup} K_1$ .

Suppose instead that the component has seven vertices. No vertex has degree at least five: its independent neighbourhood would contain a vertex of degree at most two. Thus every degree is three or four. The degree sum is even, so some vertex  $z$  has degree four. Its four neighbours are independent. Each is adjacent to both vertices outside  $N_F[z]$  to have degree at least three, and those two outside vertices are nonadjacent by triangle-freeness. Hence  $F \cong K_{3,4}$ . Taking complements leaves  $K_4 \dot{\cup} K_3$  or  $K_1 \vee (K_3 \dot{\cup} K_3)$  for  $H$ . Mader's theorem makes  $G$  seven-connected, so Corollary 2.2 gives at most one  $K_5$  subgraph. The second possibility would place  $v$  in two distinct  $K_5$  subgraphs, leaving the claimed neighbourhood.  $\square$

#### 4. ROOTED $K_{4,2}^*$ CLOSURE

For a graph  $F$  and  $S \subseteq V(F)$ , say that  $(F, S)$  is *internally  $k$ -connected* if there is no separation  $(A, B)$  of  $F$  with  $S \subseteq A$ ,  $B \setminus A \neq \emptyset$ , and  $|A \cap B| < k$ . For a four-set  $Z$ , a  $Z$ -rooted  $K_{4,2}^*$  model consists of six pairwise disjoint

connected sets: four rooted at  $Z$  and two others  $U, V$ , each adjacent to the other five. Adjacencies among the rooted sets are not required.

**Theorem 4.1** (Norin and Totschnig [10, Lemma 12]). *Let  $F$  be a graph and let  $Z \subseteq V(F)$  have order four. If  $(F, Z)$  is internally four-connected and  $F$  has no  $Z$ -rooted  $K_{4,2}^*$  model, then*

$$(4.1) \quad |E(F)| \leq 4|V(F)| - 10.$$

The following placement lemma puts one prescribed vertex in a non-root branch set.

**Lemma 4.2.** *Let  $F$  be a graph and let  $S = Z \dot{\cup} \{x\}$ , where  $|Z| = 4$ . Suppose that  $(F, S)$  is internally five-connected. If  $F$  has a  $Z$ -rooted  $K_{4,2}^*$  model, then it has one in which  $x$  belongs to a non-root branch set.*

*Proof.* Choose a  $Z$ -rooted model that maximises  $|U| + |V|$  over its two non-root branch sets  $U, V$  and, subject to that, minimises  $\sum_{i=1}^4 |R_i|$  over its rooted branch sets  $R_1, \dots, R_4$ . Write  $z_i$  for the root in  $R_i$ , and put

$$P_i = \{r \in R_i : r \text{ has a neighbour in } U \cup V\}.$$

We claim that  $|P_i| = 1$ . Both  $U$  and  $V$  are adjacent to  $R_i$ , so  $P_i$  is nonempty. If it had at least two vertices, there would be distinct  $u, w \in R_i$  such that  $u$  has a neighbour in  $U$  and  $w$  has a neighbour in  $V$ ; otherwise the two contact sets would be the same singleton. Take a minimal tree in  $F[R_i]$  containing  $z_i, u, w$ . By the minimal choice of the rooted branch sets, this tree spans  $R_i$ . One of  $u, w$ , say  $u$ , is a leaf different from  $z_i$ . Move  $u$  from  $R_i$  into  $U$ . Both altered branch sets remain connected. The former tree edge from  $u$  to  $R_i \setminus \{u\}$  preserves the contact between  $R_i$  and  $U$ , while  $w$  preserves the contact between  $R_i$  and  $V$ . Every other required adjacency is unchanged, contrary to the choice of the model. Hence  $|P_i| = 1$ .

No component outside the six branch sets can have a neighbour in  $U \cup V$ , since it could be absorbed into the adjacent branch set. It follows that

$$(4.2) \quad |N_F(U \cup V) \setminus (U \cup V)| \leq 4,$$

with at most one external neighbour in each rooted branch set. If  $x \notin U \cup V$ , (4.2) would give a separation of  $(F, S)$  of order at most four whose second open side is the nonempty set  $U \cup V$ . This contradicts internal five-connectivity. Thus  $x \in U \cup V$ .  $\square$

*Proof of Theorem 1.4.* Choose a four-set  $Z \subseteq N_G(v)$  inducing  $K_4$ , choose  $x \in N_G(v) \setminus Z$ , and put  $F = G - v$ . Deleting one vertex from a six-connected graph leaves a five-connected graph. In particular,  $(F, Z)$  is internally four-connected and  $(F, Z \cup \{x\})$  is internally five-connected. Moreover,

$$(4.3) \quad \begin{aligned} |E(F)| &= |E(G)| - d \\ &\geq 4n - 13 = 4|V(F)| - 9. \end{aligned}$$

Theorem 4.1 supplies a  $Z$ -rooted  $K_{4,2}^*$  model in  $F$ , and Lemma 4.2 lets us choose it with  $x$  in one non-root branch set.

The four rooted branch sets are pairwise adjacent through the edges of  $G[Z] = K_4$ . Together with the two non-root branch sets they form a  $K_6$  model. The singleton branch set  $\{v\}$  is adjacent to all four rooted branch sets and to the one containing  $x$ ; it may miss only the other. These seven branch sets form a  $K_7^-$  model in  $G$ .  $\square$

## 5. CONSEQUENCES FOR A CRITICAL GRAPH

Jakobsen proved that an  $n$ -vertex graph with  $n \geq 7$  and at least  $\frac{9}{2}n - 12$  edges contains a  $K_7^-$  minor or is a  $(K_{2,2,2,2}, K_6, 4)$ -cockade [6, Theorem 4]; see also [1, Theorem 2].

Write  $D(G) = 9|V(G)| - 2|E(G)|$  for the *defect* of  $G$ . The *codegree*  $c(xy) = |N(x) \cap N(y)|$  counts the triangles containing an edge  $xy$ . Contracting an edge reduces connectivity by at most one: a cut avoiding the contraction vertex lifts unchanged, while a cut containing it lifts after that vertex is replaced by the two ends. For an edge  $e$  of codegree  $c$ ,

$$(5.1) \quad |E(G/e)| = |E(G)| - 1 - c, \quad D(G/e) = D(G) - 7 + 2c.$$

**Lemma 5.1.** *Every five-connected  $K_7^-$ -minor-free graph  $G$  of order at least nine satisfies  $D(G) \geq 25$ .*

*Proof.* The order excludes the two base cockades  $K_6$  and  $K_{2,2,2,2}$ . A nontrivial cockade has a four-vertex cut, contrary to five-connectivity. Jakobsen's strict inequality  $2|E(G)| < 9|V(G)| - 24$  and integrality give  $D(G) \geq 25$ .  $\square$

**Proposition 5.2.** *Every graph satisfying the hypotheses of Theorem 1.1 is seven-connected, has minimum degree eight, has no  $K_5$  subgraph, and satisfies  $|E(G)| \geq 4|V(G)|$  and  $n_8 \geq 25 + \tau$ .*

*Proof.* Albar's theorem gives  $\chi(G) = 7$ , and Mader's theorem gives seven-connectivity. By Theorem 3.2 and Corollary 2.2, every degree-seven vertex lies in the unique possible  $K_5$ , so  $n_7 \leq 5$ . Degree summation gives  $2|E(G)| \geq 8|V(G)| - 5$  and hence  $|E(G)| \geq 4|V(G)| - 2$ . If  $v$  had degree seven, its neighbourhood would contain a  $K_4$  and this density would exceed the threshold  $4|V(G)| - 6$  in Theorem 1.4. Thus no such vertex exists:  $\delta(G) \geq 8$  and  $|E(G)| \geq 4|V(G)|$ .

Put  $q = |E(G)| - 4|V(G)|$ , so  $q \geq 0$ . Suppose that a  $K_5$  subgraph has vertex set  $K$ . For each  $w \in K$ , the neighbourhood of  $w$  contains the  $K_4$  on  $K \setminus \{w\}$ . Since  $G$  has no  $K_7^-$  minor, the contrapositive of Theorem 1.4 gives

$$4|V(G)| + q < 4|V(G)| + d_G(w) - 13,$$

and hence  $d_G(w) \geq q + 14$ . The five vertices of  $K$  would contribute at least  $5(q + 6)$  to

$$\sum_{z \in V(G)} (d_G(z) - 8) = 2q,$$

which is impossible. Thus  $G$  contains no  $K_5$  subgraph.

The minimum degree ensures  $|V(G)| \geq 9$ . Lemma 5.1 and degree summation now give

$$25 \leq 9|V(G)| - 2|E(G)| = n_8 - \tau.$$

Therefore  $n_8 \geq 25 + \tau$ .  $\square$

## 6. DEGREE-EIGHT VERTICES WITHOUT COMPUTATION

**Lemma 6.1.** *Let  $G$  satisfy the hypotheses of Theorem 1.1. Every degree-eight vertex  $v$  satisfies  $\alpha(G[N(v)]) = 3$ .*

*Proof.* Dirac's inequality (3.1) gives the upper bound three. Put  $J = G[N(v)]$ . By Proposition 5.2,  $J$  has no  $K_4$  subgraph and  $|V(G)| \geq n_8 \geq 25$ , so  $G - N[v]$  is nonempty. If  $\alpha(J) \leq 2$ , Rolek, Song and Thomas [13, Lemma 2.1] show that  $J$  contains a spanning square of an eight-cycle. Label its vertices cyclically  $0, \dots, 7$ . An exterior component has at least seven neighbours in  $N(v)$ , by seven-connectivity. Contract that component to  $c$  and delete the other exterior vertices. By cyclic symmetry and edge deletion, take  $c$  adjacent to every cycle vertex except 0. The sets

$$\{0, 7, 2\}, \quad \{3\}, \quad \{4\}, \quad \{1, v\}, \quad \{6\}, \quad \{5\}, \quad \{c\}$$

are connected and pairwise adjacent except for  $\{3\} : \{6\}$ . They form a forbidden  $K_7^-$  model. Hence  $\alpha(J) = 3$ .  $\square$

**Lemma 6.2.** *Let  $J$  be an eight-vertex graph with minimum degree at least four, no  $K_4$  subgraph, and independence number three. Call  $r$  good if  $J - r$  has a  $K_5^-$  minor. Every vertex of  $J$  has a good neighbour.*

*Proof.* The graph  $J$  is three-connected. Indeed, a component has order at least five, and a component behind a cutvertex has order at least four, excluding both kinds of separation on eight vertices. A two-cut would leave two three-vertex components, each a triangle complete to the cut. An edge in the cut gives a  $K_4$ ; an independent cut gives independence number two. Both are impossible.

Suppose first that  $J$  is four-connected. Then  $J - r$  is three-connected for every  $r$ . Wood and Woodall [14, Lemma 4.2.1] classify three-connected  $K_5^-$ -minor-free graphs as wheels, the triangular prism and  $K_{3,3}$ . If some  $r$  is not good, order seven forces  $J - r$  to be a wheel with six rim vertices. Every rim vertex must see  $r$  to have degree at least four, and the hub cannot see  $r$  because  $J$  has no  $K_4$ . Thus  $J = \overline{K_2} \vee C_6$ . With apices  $p, q$  and rim  $v_0, \dots, v_5$ , deletion of  $v_2$  leaves the  $K_5^-$  model

$$\{p, v_0\}, \quad \{q, v_1\}, \quad \{v_3\}, \quad \{v_4\}, \quad \{v_5\},$$

missing only  $\{v_3\} : \{v_5\}$ . Every rim vertex is therefore good, and the rim vertices meet every neighbourhood. If no bad  $r$  exists, the conclusion is immediate.

Now let  $S$  be a three-cut. Its two components have orders two and three, since each has order at least two. Write the first as  $A = \{a, a'\}$  and the second as  $B$ . The edge  $aa'$  is present and both ends are complete to  $S$ .

Consequently  $S$  is independent. Every vertex of  $B$  has a neighbour in  $B$ , so  $J[B]$  is a path or triangle.

If  $B = b_1 b_2 b_3$  is a path, its ends are complete to  $S$  and  $b_2$  has at least two neighbours in  $S$ . The following models establish that all vertices in  $A \cup B$  are good; symmetry handles  $a'$  and  $b_3$ . In the table,  $b_1 s_1$  abbreviates the branch set  $\{b_1, s_1\}$ , and similarly for the other entries.

| deleted vertex | five branch sets                 | only missing pair |
|----------------|----------------------------------|-------------------|
| $a$            | $b_1 s_1, b_3 s_2, b_2, a', s_3$ | $b_2 : a'$        |
| $b_1$          | $a s_1, b_3 s_2, b_2, a', s_3$   | $b_2 : a'$        |
| $b_2$          | $a s_1, b_1 s_2, b_3, a', s_3$   | $b_3 : a'$        |

Here  $S = \{s_1, s_2, s_3\}$  is labelled with  $b_2 s_3$  present in the first row, and with  $b_2 s_1, b_2 s_3$  present in the second.

If  $B$  is a triangle, each of its vertices sees at least two vertices of  $S$ , and no vertex of  $S$  sees all of  $B$ . Its missing incidences with  $S$  therefore form a perfect matching; label these  $b_i s_i$ . After deleting  $a$  use

$$s_1 b_2, \quad s_2 a', \quad b_1, \quad s_3, \quad b_3,$$

which misses only  $s_3 : b_3$ ; after deleting  $b_1$  use

$$s_1 b_2 b_3, \quad s_2, \quad s_3, \quad a, \quad a',$$

which misses only  $s_2 : s_3$ . Symmetry again makes all vertices in  $A \cup B$  good. In either case, the vertices of  $A$  see one another, every vertex of  $B$  has a neighbour in  $B$ , and every vertex of  $S$  sees  $A$ . Thus every vertex has a good neighbour.  $\square$

**Proposition 6.3.** *Let  $G$  satisfy the hypotheses of Theorem 1.1. Every degree-eight vertex is incident with an edge  $e$  satisfying  $c(e) \leq 3$ , and  $n_8 \geq 26 + \tau$ . Both conclusions have computation-free proofs.*

*Proof.* Fix a degree-eight vertex  $v$  and put  $J = G[N(v)]$ . By Proposition 5.2 and Lemma 6.1,  $J$  is  $K_4$ -free and has independence number three. If every incident edge has codegree at least four, then  $\delta(J) \geq 4$ . An exterior component exists because  $|V(G)| \geq 25$ , and seven-connectivity gives at least seven boundary neighbours. Contract it to  $c$  and delete the other exterior components. Let  $x$  be the vertex of  $J$  missed by  $c$ , or any vertex if none is missed. By Lemma 6.2,  $x$  has a neighbour  $r$  such that  $J - r$  has a  $K_5^-$  model  $M_1, \dots, M_5$ . Then  $cr$  is an edge, and

$$\{v\}, \quad \{c, r\}, \quad M_1, \dots, M_5$$

form a  $K_7^-$  model: the first two sets are adjacent through  $vr$  and are adjacent to every  $M_i$ . The sole possible missing contact from  $c$  is a singleton  $M_i = \{x\}$ , supplied instead by  $rx$ . This contradiction proves that some edge  $e = vx$  has  $c(e) = d_J(x) \leq 3$ .

The quotient  $G/e$  is six-connected, has no  $K_7^-$  minor and has order at least twenty-four. Lemma 5.1 gives  $D(G/e) \geq 25$ , while (5.1) gives  $D(G) \geq D(G/e) + 1 \geq 26$ . The identity  $D(G) = n_8 - \tau$  proves the bound.  $\square$

## 7. CONNECTIVITY AND DEFECT

**Lemma 7.1.** *Every eight-vertex graph  $J$  of minimum degree at least five contains a  $K_6^-$  minor.*

*Proof.* The complement of  $J$  has maximum degree at most two. Extend it to an edge-maximal graph  $F$  with that property. Every component of  $F$  is a cycle, except possibly one component which is  $K_1$  or  $K_2$ : any longer path could be closed, and deficient vertices in two components could be joined. The seven possibilities for  $F$  and explicit models in its complement are displayed below. Cycle labels are consecutive; a string such as 03 denotes the set  $\{0, 3\}$ .

| $F$                                                          | six branch sets    | only missing pair |
|--------------------------------------------------------------|--------------------|-------------------|
| $C_8(0, \dots, 7)$                                           | 03, 15, 2, 4, 6, 7 | 6 : 7             |
| $C_5(0, \dots, 4) \dot{\cup} C_3(5, 6, 7)$                   | 05, 1, 26, 3, 4, 7 | 3 : 4             |
| $C_4(0, \dots, 3) \dot{\cup} C_4(4, \dots, 7)$               | 04, 1, 25, 3, 6, 7 | 6 : 7             |
| $C_7(0, \dots, 6) \dot{\cup} K_1(7)$                         | 024, 1, 3, 5, 6, 7 | 5 : 6             |
| $C_4(0, \dots, 3) \dot{\cup} C_3(4, 5, 6) \dot{\cup} K_1(7)$ | 024, 1, 3, 5, 6, 7 | 5 : 6             |
| $C_6(0, \dots, 5) \dot{\cup} K_2(6, 7)$                      | 024, 1, 3, 5, 6, 7 | 6 : 7             |
| $C_3(0, 1, 2) \dot{\cup} C_3(3, 4, 5) \dot{\cup} K_2(6, 7)$  | 03, 14, 2, 5, 6, 7 | 6 : 7             |

Each nonsingleton is connected in  $\overline{F}$ , and all interbag adjacencies except the indicated one are present. Since  $\overline{F}$  is a spanning subgraph of  $J$ , each displayed model also lies in  $J$ .  $\square$

Theorem 1.2 needs an edge of codegree at most three without a colouring hypothesis. Contracting exterior components reduces the degree-seven and degree-eight cases to the following finite lemma.

**Lemma 7.2** (Computer-assisted finite lemma). *For a graph  $J$  and  $A \subseteq V(J)$ , let  $Q(J, A)$  be obtained by adding nonadjacent vertices  $v, c$ , making  $v$  complete to  $J$  and giving  $c$  precisely the neighbourhood  $A$ .*

- (1) *If  $|V(J)| = 7$ ,  $\delta(J) \geq 4$  and  $|A| \geq 6$ , then  $K_7^- \preceq Q(J, A)$ .*
- (2) *Fix a labelled graph  $J$  with  $|V(J)| = 8$ ,  $\delta(J) \geq 4$  and  $K_6^- \not\preceq J$ . At most one set  $A \subseteq V(J)$  with  $|A| \geq 6$  makes  $Q(J, A)$   $K_7^-$ -minor-free. If such a set exists,  $V(J) \setminus A$  consists of two adjacent vertices of degree four in  $J$ .*

*Finite verification and its scope.* For part (1),  $\overline{J}$  has maximum degree two, hence is a union of paths and cycles. Enumerating the multisets of such components with total order seven gives 29 types. For each, check the full attachment and its seven possible single omissions. Each of the  $29 \cdot 8 = 232$  nine-vertex quotients has a seven-bag model. The checker enumerates all 750 partitions of vertex subsets into seven nonempty bags, and tests connectivity and all interbag adjacencies. A seven-bag model on nine vertices uses seven, eight or nine vertices, so this search is exhaustive.

For part (2), adjoin a vertex with each possible neighbourhood to each of the 1,044 unlabelled seven-vertex graphs in the NetworkX graph atlas.

Every eight-vertex graph occurs. The minimum-degree filter and exact isomorphism testing leave 424 classes. Of these, 55 have no  $K_6^-$  minor. Checking all missed sets of order zero, one or two gives  $55(1 + 8 + 28) = 2,035$  exterior profiles. Exactly 2,031 have certified  $K_7^-$  models. Each of the four remaining profiles has the uniqueness and degree properties in part (2).

Here the minor engine starts with singleton bags and recursively either deletes a bag or merges two touching bags. These operations preserve disjoint connected bags and generate every minor model by contracting spanning trees and deleting unused vertices. At six or seven bags it accepts precisely when at most one pair is nonadjacent. Every positive certificate is separately checked for disjointness, connectivity and interbag adjacencies. The implementations test known positive and negative instances; a separate internal audit also checked the four negative quotients by all 11,880 seven-bag partitions of their vertex subsets.

The accompanying reproducibility note, `README.md`, records the four profiles, executable inputs and SHA-256 certificate digests. Part (1) uses only the Python standard library; part (2) also uses NetworkX's atlas and exact isomorphism implementation. The reductions below impose no bound on the order of an exterior component or of  $G$ .  $\square$

**Theorem 7.3.** *In every six-connected  $K_7^-$ -minor-free graph, each degree-eight vertex is incident with an edge of codegree at most three.*

*Proof.* Let  $v$  have degree eight and put  $J = G[N(v)]$ . Suppose every incident codegree is at least four. Then  $\delta(J) \geq 4$ , and  $J$  has no  $K_6^-$  minor because adjoining  $\{v\}$  would give the forbidden minor. If  $G - N[v]$  is empty, minimum degree six in  $G$  gives  $\delta(J) \geq 5$ , contradicting Lemma 7.1.

Every component  $C$  of  $G - N[v]$  has at least six neighbours in  $N(v)$ , since its neighbourhood separates it from  $v$ . Contracting  $C$  to  $c$  and deleting the other exterior components gives exactly  $Q(J, N_G(C))$ . Lemma 7.2(2) therefore applies to every exterior component. For the fixed labelled graph  $J$ , all components must miss the same unique pair  $a, b$ . Neither  $a$  nor  $b$  has an exterior neighbour, so both have degree  $1 + 4 = 5$  in  $G$ , contradicting six-connectivity.  $\square$

The degree-six case adapts a computation-free argument of Norin and Totschnig. We verify the hypotheses under  $K_7^-$  exclusion.

**Lemma 7.4** (Degree-six disc bound). *Let  $H$  be a five-connected  $K_7^-$ -minor-free graph on at least nine vertices. If every edge has codegree at least four and some vertex has degree six, then  $|E(H)| \leq 4|V(H)| - 9$ .*

*Proof.* Let  $v$  have degree six. Since  $d_{H[N(v)]}(x) = c(vx) \geq 4$ , label  $N(v) = \{u_1, w_1, u_2, w_2, u_3, w_3\}$  so that all its possible nonedges are  $u_i w_i$ . Two disjoint paths joining two distinct displayed pairs, with open interiors outside  $N[v]$ , would fill two of the three missing edges after contraction. Together with  $v$  they give a  $K_7^-$  model. We call this the two-pair obstruction; a present pair-edge may serve as one of the paths.

All three displayed pair-edges are absent. Otherwise an exterior component, with at least five neighbours in  $N(v)$  by five-connectivity, joins a second pair and violates the two-pair obstruction. Moreover, some pair, say  $u_1, w_1$ , has no exterior common neighbour. Otherwise choose a common neighbour for each pair. Two distinct choices contradict the obstruction. If all choices coincide at  $x$ , order at least nine leaves a component outside  $N[v] \cup \{x\}$ . Its boundary has at least five vertices, at least four of which belong to  $N(v)$ . It therefore joins one pair disjointly from a second pair's path through  $x$ , again a contradiction.

We now use the two-paths theorem in the form of [10, Theorem 13]. The next two applications are the degree-six argument in [10, Claims 3.12–3.15]; we record the separator and density hypotheses needed for them. There is no nontrivial separation  $(A, B)$  of order five with  $N[v] \subseteq A$ . Indeed, five-connectivity and Menger's theorem provide five disjoint paths in  $H[A]$  from five distinct members of  $N(v)$  to the five boundary vertices, with interiors avoiding  $N[v]$ . These five members contain two complete matching pairs. Temporarily label their boundary ends  $s_1, t_1$  and  $s_2, t_2$ , and label the fifth end  $x$ . Apply the two-paths theorem in  $H[B] - x$  to the order  $s_1, s_2, t_1, t_2$ . A crossing violates the two-pair obstruction. A separation outcome lifts after restoring  $x$  to a cut of order at most four in  $H$ . In the remaining outcome the graph embeds in a disc with the four terminals on its boundary. An edge with an end in  $B \setminus A$  has at most two common neighbours in that disc: three would enclose a vertex behind a triangle, which together with  $x$  gives a cut of order at most four in  $H$ . Restoring  $x$  gives codegree at most three, a contradiction.

Apply the same theorem to  $H' = H - \{v, u_1, w_1\}$ , with terminal order  $u_2, u_3, w_2, w_3$ . A crossing again violates the two-pair obstruction. A separating outcome lifts, after restoring  $u_1, w_1$ , to a separation of order at most five with  $N[v]$  on one side; five-connectivity and the preceding paragraph exclude it. Thus  $H'$  embeds in a disc with the four terminals on its boundary. Put  $n' = |V(H')|$ . Then  $|E(H')| \leq 3n' - 7$ . Each of its vertices outside  $N[v]$  sees at most one of  $\{v, u_1, w_1\}$ ; the four remaining neighbourhood vertices see all three, and the deleted triple induces two edges. Hence

$$|E(H)| \leq (3n' - 7) + (n' - 4) + 12 + 2 = 4n' + 3 = 4|V(H)| - 9.$$

The forbidden-minor hypothesis was used only for the displayed  $K_7^-$  two-pair model. Neither edge-maximality nor the stronger  $K_7^\vee$  exclusion is used in this argument.  $\square$

**Proposition 7.5.** *Let  $G$  be six-connected and  $K_7^-$ -minor-free. If  $|E(G)| \geq 4|V(G)|$ , then some edge of  $G$  has codegree at most three.*

*Proof.* The density implies order at least nine, so Lemma 5.1 gives average degree less than nine. Thus a vertex of minimum degree has degree six, seven or eight. Suppose every edge has codegree at least four. Lemma 7.4

excludes degree six, and Theorem 7.3 excludes degree eight. At a degree-seven vertex  $v$ , the local graph  $J = G[N(v)]$  has minimum degree at least four. There is an exterior component, and six-connectivity gives it at least six boundary neighbours. Contracting it produces a quotient forbidden by Lemma 7.2(1). This excludes all three degrees.  $\square$

*Proof of Theorem 1.2.* Let  $G$  satisfy the hypotheses. Proposition 7.5 supplies an edge  $e$  of codegree at most three. The quotient  $G/e$  is  $(r - 1)$ -connected and has no  $K_7^-$  minor. By (5.1),  $|E(G/e)| \geq 4|V(G/e)|$  and  $D(G) \geq D(G/e) + 1$ . The retained density ensures  $|V(G/e)| \geq 9$ .

For  $r = 6$ , Lemma 5.1 gives  $D(G/e) \geq 25$ , so  $D(G) \geq 26$ . For  $r \geq 7$ , induction on  $r$  applies to  $G/e$  and gives

$$D(G) \geq D(G/e) + 1 \geq 20 + (r - 1) + 1 = 20 + r.$$

$\square$

The base case uses Jakobsen's bound because contraction need not preserve six-connectivity.

## 8. SEVEN-VERTEX SEPARATORS

This section is computation-free. A connected subgraph of  $G - S$  is *full at  $S$*  if it is adjacent to every vertex of  $S$ . We retain the exact equality partition of a boundary colouring, because its mere number of colours does not suffice for gluing.

**Lemma 8.1.** *Let  $V(G) = L \dot{\cup} S \dot{\cup} R$ , with  $L, R$  nonempty and anticomplete, and suppose every proper minor of  $G$  is six-colourable. Let  $\Pi$  partition  $S$  into independent blocks. Suppose that all its blocks, except some singleton blocks whose vertices form a clique, can be assigned injectively to disjoint connected subgraphs of  $G[L]$  full at  $S$ , with at least one block assigned. Then  $G[R \cup S]$  has a six-colouring with equality partition on  $S$  exactly  $\Pi$ .*

*Proof.* For each assigned block  $B_j$  and its full subgraph  $P_j$ , contract the connected set  $B_j \cup V(P_j)$ . These sets are disjoint. Their images, together with the retained singleton blocks, form a clique: fullness supplies all adjacencies involving an assigned block, and the retained singletons form a clique by hypothesis. At least one edge is contracted, so the resulting minor is proper. Six-colour it and pull back only to  $R \cup S$ , assigning a block the colour of its image. Each block is independent and every edge to a retained vertex was represented in the minor, so the pullback is proper. The clique forces distinct colours on distinct blocks, giving exactly  $\Pi$ .  $\square$

**Lemma 8.2.** *A graph on seven vertices with at least ten edges contains a  $K_4^-$ , a house, or  $K_{2,3}$  as a not necessarily induced subgraph. A house is a four-cycle with an additional vertex joined to the ends of one cycle edge.*

*Proof.* Suppose  $Q$  contains none of the three. Choose a maximum-degree vertex  $v$ , put  $d = d(v)$ ,  $A = N(v)$ , and  $B = V(Q) \setminus N[v]$ . The graph  $Q[A]$

is a matching, since a two-edge path there gives  $K_4^-$  with  $v$ . Every vertex of  $B$  has at most two neighbours in  $A$ , since three would give  $K_{2,3}$  with  $v$ . For  $d \geq 5$ ,

$$|E(Q)| \leq d + \lfloor d/2 \rfloor + 2(6-d) + \binom{6-d}{2} \leq 9.$$

For  $d \leq 2$  there are at most seven edges. Write  $\alpha = |E(Q[A])|$ ,  $x = |E_Q(A, B)|$ ,  $\beta = |E(Q[B])|$ .

If  $d = 4$ , then  $\alpha \leq 2$ ,  $x \leq 4$ ,  $\beta \leq 1$ . For  $\alpha = 2$ , ten edges force a vertex of  $B$  with two neighbours in  $A$ . If these are matched, there is a  $K_4^-$ ; if they are on distinct matching edges, there is a house. For  $\alpha \leq 1$ , ten edges force  $(\alpha, x, \beta) = (1, 4, 1)$ . Let  $ab$  be the edge in  $A$  and  $c, t$  its isolated vertices. A two-set of neighbours in  $A$  using  $a$  or  $b$  gives  $K_4^-$  or a house. Thus the two vertices  $p, q$  of  $B$  both see  $c, t$ ; their edge gives a  $K_4^-$  on  $p, q, c, t$ .

If  $d = 3$ , maximum degree gives exactly ten edges, so  $\alpha + x + \beta = 7$  and  $\alpha \leq 1$ . For  $\alpha = 1$ , no vertex of  $B$  can have two neighbours in  $A$ , by the same house or  $K_4^-$  argument. Hence  $x = \beta = 3$ :  $B$  is a triangle and its three edges to  $A$  have distinct ends, since a repetition gives  $K_4^-$ . The edge of  $A$ , its two matched vertices in  $B$ , and  $v$  form a house.

It remains that  $\alpha = 0$ . If  $\beta = 3$ , each vertex of  $B$  already has two neighbours in  $B$ , so  $x \leq 3$ , a contradiction. If  $\beta = 2$ , write  $B = pqr$  as a path. Then  $x = 5$ , and its cross-degrees are 2, 1, 2. Equal two-element  $A$ -neighbourhoods of  $p, r$  give  $K_{2,3}$ . If they differ, the neighbour of  $q$  is either their common vertex, giving  $K_4^-$ , or one of the two other vertices, giving a house. If  $\beta = 1$ , every vertex of  $B$  has two neighbours in  $A$ . Repeated two-sets give  $K_4^-$  or  $K_{2,3}$ . Otherwise label the edge of  $B$  as  $pq$  and write

$$N_A(p) = \{a, b\}, \quad N_A(q) = \{a, c\}, \quad N_A(r) = \{b, c\}.$$

Then  $a, v, b, p$  is a four-cycle with roof  $q$ . Finally  $\beta = 0$  would force  $x = 7 > 6$ . Thus  $Q$  has at most nine edges.  $\square$

**Lemma 8.3.** *Let  $G$  satisfy the hypotheses of Theorem 1.1, and let  $S$  be a seven-vertex cut. Then  $G - S$  has two or three components. If it has three,  $G[S]$  has a proper three-colouring and every such colouring has class sizes 3, 2, 2.*

*Proof.* Every component is full at  $S$  by seven-connectivity. If there are at least five, contract five of them to  $c_1, \dots, c_5$ , label  $S = \{s_1, \dots, s_7\}$ , and use

$$\{c_i, s_i\} \ (1 \leq i \leq 5), \quad \{s_6\}, \{s_7\}.$$

All but possibly the last pair are adjacent, giving  $K_7^-$ .

Suppose there are four components, contracted to  $c_1, \dots, c_4$ . If  $G[S]$  has a path  $xyz$ , choose three further vertices  $u, v, w$  of  $S$  and use

$$\{c_1\}, \quad \{c_2, u\}, \quad \{c_3, v\}, \quad \{c_4, w\}, \quad \{x\}, \quad \{y\}, \quad \{z\}.$$

Only  $\{x\} : \{z\}$  may be nonadjacent, again impossible. Hence  $G[S]$  is a matching and isolated vertices. It has an isolated vertex  $q$  and a partition

$S = I_1 \dot{\cup} I_2 \dot{\cup} \{q\}$  into independent sets of orders 3, 3, 1. Split the four components into two pairs, whose unions are  $L, R$ . In each union the two full components can be assigned to  $I_1, I_2$  in Lemma 8.1, retaining  $q$ . The two colourings have the same exact boundary partition; permuting colour names makes them agree on  $S$  and yields a six-colouring of  $G$ , a contradiction. Thus there are at most three components.

Suppose there are three, with contracted vertices  $c_1, c_2, c_3$ . If  $G[S]$  had at least ten edges, Lemma 8.2 would supply one of its three subgraphs. For a  $K_4^-$  subgraph, retain its four vertices and merge each  $c_i$  with one of the three unused boundary vertices. For a house or  $K_{2,3}$ , contract respectively the square edge opposite the roof or any edge to obtain a four-bag  $K_4^-$  model using five boundary vertices. Retain  $c_1$  and merge  $c_2, c_3$  with the two unused boundary vertices. In either case the seven sets form a  $K_7^-$  model. Consequently  $|E(G[S])| \leq 9$ . The boundary also has no  $K_4$  subgraph, by the same construction using four singleton roots.

Every  $K_4$ -free graph on seven vertices with at most nine edges is three-colourable. Otherwise take a four-critical subgraph  $F$ ; it has minimum degree at least three. On four vertices it is  $K_4$ . On five vertices parity gives a universal vertex, whose deletion leaves a three-chromatic four-vertex graph, containing a triangle; this also gives  $K_4$ . On six vertices a degree at least four forces ten edges. Otherwise  $F$  is cubic, with complement  $C_6$  or  $2C_3$ ; pairing consecutive vertices in the former or taking the triangles in the latter gives a three-colouring. On seven vertices minimum degree three gives at least eleven edges. These contradictions prove the assertion.

If  $G[S]$  were two-colourable, fix its partition into at most two independent blocks. For each component use the other two full components in Lemma 8.1 to reproduce the partition on that closed component. The three colourings align and glue, a contradiction. Likewise, if any proper three-colouring has a singleton class, retain that singleton and assign the other two blocks to the other two components. The same gluing contradiction results. Every class of a proper three-colouring therefore has order at least two; their orders are 3, 2, 2.  $\square$

**Theorem 8.4.** *Let  $G$  be seven-connected, with every proper minor six-colourable. Let  $S$  be a seven-vertex cut with exactly three components  $D_1, D_2, D_3$ . If  $G[S]$  has a proper colouring with class sizes 3, 2, 2, then  $G$  is six-colourable. No excluded-minor hypothesis is needed.*

*Proof.* Fix the independent boundary blocks  $S = T \dot{\cup} A \dot{\cup} B$ , with  $|T| = 3$  and  $|A| = |B| = 2$ . Every  $D_i$  is full at  $S$ . Say that a component realises  $(A, B)$  if it contains disjoint connected subgraphs  $X_A, X_B$ , the first adjacent to both vertices of  $A$ , the second to both vertices of  $B$ . They can also be made adjacent: join them by a shortest path in the component, split it at an edge, and absorb the two halves into the respective subgraphs.

If two components realise  $(A, B)$ , then for every retained component  $D_i$  we may choose a different realising component  $R$  and let  $F$  be the third.

Contract the three disjoint connected sets

$$X_A \cup A, \quad X_B \cup B, \quad F \cup T.$$

Their images form a triangle, using the adjacency of  $X_A$  and  $X_B$  and fullness of  $F$ . A six-colouring of this proper minor pulls back on  $G[D_i \cup S]$  with exact boundary partition  $T|A|B$ . The three colourings then align and glue. Thus we may assume  $D_1, D_2$  do not realise  $(A, B)$ .

Fix one of these two components  $D_i$ . Add four artificial terminals  $a_1^*, b_1^*, a_2^*, b_2^*$ , each with the neighbours in  $D_i$  of its corresponding literal root. There are no two disjoint paths joining the  $a$ -pair and the  $b$ -pair, since their interiors would realise  $(A, B)$ . Add the cycle edges in the displayed terminal order. These do not create such a crossing: a path using an edge between consecutive terminals would meet a terminal of the other prescribed path. Extend on the same vertex set to an edge-maximal graph with no such crossing. Humeau and Pous [5, Theorem 1.3] identify this completion as a web with the four terminals as frame. In this form a web consists of a planar graph with the frame on a face and with cliques attached behind facial triangles.

No clique inserted behind such a triangle can contain a vertex of  $D_i$  outside the triangle. Otherwise those original vertices have no auxiliary neighbours outside the triangle. Replace any artificial terminal of the triangle by its literal root and add the three omitted vertices of  $T$ . These at most six actual vertices then separate a nonempty part of  $D_i$  from another component of  $G - S$ , contrary to seven-connectivity. Thus every original vertex of  $D_i$  belongs to the planar part of the web. Replacing the artificial terminals by their literal roots shows that

$$H_i^+ = G[D_i \cup A \cup B] + E(K_{A,B})$$

is planar, with the added  $K_{2,2}$  as an induced facial four-cycle. The cycle is induced because  $A$  and  $B$  are independent. These added edges serve only in the planar extension; they are not asserted to be edges of  $G$  or used in a minor of  $G$ .

Contract in  $G$  the two disjoint connected sets

$$Q_0 = D_1 \cup T, \quad Q_1 = D_2 \cup A.$$

Their images are adjacent and each sees both retained vertices of  $B$ . Six-colour this proper minor. Pulling back on  $G[D_3 \cup S]$  gives a colouring  $c$  in which  $T$  has colour 0,  $A$  a different colour 1, and both vertices of  $B$  avoid 0, 1.

For  $i = 1, 2$ , precolour the induced four-cycle of  $H_i^+$  by  $c|_{A \cup B}$ . It uses at most three colours from the five-colour palette excluding 0. Diwan [3, Corollary 1] proves that a proper precolouring of an induced cycle of length at most  $2k - 5$  using at most  $k - 1$  colours extends to a  $k$ -colouring of every planar supergraph containing it. With  $k = 5$  it extends to each  $H_i^+$ . Use these extensions on  $D_1, D_2$  and retain  $c$  on  $D_3 \cup S$ . They agree on  $A \cup B$

and avoid 0 on  $D_1, D_2$ , so every edge to  $T$  is proper. Distinct components are anticomplete. The combined colouring is a six-colouring of  $G$ .  $\square$

*Proof of Theorem 1.1.* Proposition 5.2 gives seven-connectivity, minimum degree eight,  $|E(G)| \geq 4|V(G)|$  and the absence of a  $K_5$  subgraph. The last conclusion makes every neighbourhood  $K_4$ -free. Lemma 6.1 gives independence number three in every degree-eight neighbourhood; Proposition 6.3 gives the incident-edge conclusion and  $n_8 \geq 26 + \tau$  without computation. Theorem 1.2 applied with  $r = 7$  gives  $9|V(G)| - 2|E(G)| \geq 27$ . The exact identity  $9|V(G)| - 2|E(G)| = n_8 - \tau$  proves the asserted stronger bound. For a seven-vertex cut  $S$ , Lemma 8.3 leaves two or three components. Three would give a 3, 2, 2 boundary colouring, and Theorem 8.4 would make  $G$  six-colourable. Thus  $G - S$  has exactly two components.  $\square$

#### STATEMENT ON AI USE AND INDEPENDENCE

OpenAI models contributed to theorem discovery, proofs, verification and drafting. Cavit Erginsoy directed the work and made the final decisions. Erginsoy is not affiliated with OpenAI, and OpenAI neither sponsored nor verified the work.

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